

Jarvin Chavez

Full Stack Software Engineer | AI Systems

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Founding software engineer specializing in production AI pipelines, scalable backend architecture, and data-intensive applications across startup, research, and enterprise environments.

EDUCATION

University of St. Thomas

Houston, TX | 2021 - 2026

B.S. in Computer Science · Dean's List (2021–2024) · Relevant Coursework: Data Science, Data Analytics, Object-Oriented Programming, Computer Architecture

EXPERIENCE

Helen · Kassandra Wellness Group

2026 - Present

Founding Software Engineer & Technical Advisor

- Architected and delivered Helen's full-stack platform, including backend services, REST APIs, database layers, and React user interfaces.
- Designed AI-powered search and recommendation systems to improve symptom analysis, provider matching, and user engagement.
- Engineered a search system indexing and ranking results across millions of MongoDB records using custom AI ranking algorithms.
- Deployed scalable cloud infrastructure and data pipelines supporting high-volume search, analytics, and production workloads.
- Advised leadership on product strategy, technical roadmap, feature prioritization, and long-term platform architecture.

Corva AI

2024 - 2025

Backend Software Engineering Intern

- Engineered Python and Node.js backend services processing 1,000+ daily messages via Redis-backed queues on AWS in production.
- Developed database-driven display pipelines over PostgreSQL and MongoDB enabling real-time AI analysis for end users.
- Integrated AI analysis pipelines and maintained RESTful APIs for data ingestion, transformation, and internal service communication.
- Deployed and monitored containerized services on AWS with Docker and GitHub Actions CI/CD pipelines.

CareerSpring

2025

Backend Developer

- Developed backend services and database-driven APIs surfacing thousands of records for business analysis and operational decision-making.
- Built Power BI integrations connecting live data sources to automated reporting dashboards for internal stakeholders.
- Improved RESTful API reliability and performance across distributed data pipeline systems.

University of St. Thomas - NSF Research

2022 - 2024

Undergraduate Researcher & Full Stack Developer

- Processed and analyzed hundreds of thousands of proteins from UniProt using async pipelines in Python, Celery, and Redis.
- Integrated UniProt API ingestion with 3D molecular visualization displays to enable interactive protein structure analysis.
- Applied scikit-learn and pandas for ML-based clustering and statistical analysis on large biological datasets.
- Presented bioinformatics research findings at ABRCMS 2023.

PROJECTS

Protein Motif Clustering Pipeline — Vectorized and clustered motif data from hundreds of thousands of UniProt proteins for unsupervised bioinformatics analysis using scikit-learn.

ηeta — Desktop Structured Notebook (Capstone) — Cross-platform desktop application in C++ and Qt 6 with modular architecture, custom expression evaluator, JSON persistence, and CMake builds for macOS, Windows, and Linux.

Crash Contributing Factors Analysis — End-to-end Python pipeline analyzing FARS national crash data to identify alcohol-related risk patterns across thousands of incidents.

iCn3D — MCD Miner Integration — Proof-of-concept integrating iCn3D 3D molecular visualization into the MCD Miner bioinformatics research pipeline.

ADDITIONAL EXPERIENCE

TG Natural Resources, IT Department · IT Intern (2023) — Data migrations, hardware deployments, and end-user technical support.

Ellucian · IT Intern (2021–2022) — System configuration, software migrations, and secured network infrastructure.

SKILLS & CERTIFICATIONS

Languages & Frameworks: Python, JavaScript, TypeScript, React, Node.js, Java, C++, HTML/CSS, FastAPI, Express.js

Cloud & DevOps: AWS EC2, GCP, Docker, Kubernetes, Linux, Git, GitHub Actions, CI/CD, Power BI

Databases: MongoDB, PostgreSQL, MySQL, SQLite, SQL Server, Redis

AI & Data: PyTorch, TensorFlow, scikit-learn, pandas, NumPy, Celery, Prompt Engineering

Security: AI/LLM Security, Prompt Injection Mitigation, Secure API Design, Network Access Control

Certifications: AWS Cloud Foundations · AWS Machine Learning Foundations · AWS Data Science · AWS Prompt Engineering

Languages Spoken: English (Native) · Spanish (Native) · French (Beginner)